

EC3213 — Money, Credit and Banking

Winter 2024 Examination · Paper 1 · Full Marking Scheme

Module	EC3213 — Money, Credit and Banking
Internal Examiner	Dr J. Gao
Institution	University College Cork
Session	Winter 2024, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions: attempt 2 of 3 (100% of the paper). All questions carry equal weight.
Materials	Non-programmable calculator permitted

Coverage note. This scheme answers **all three** Section A questions in full, since you'll choose your best two under exam conditions.

Own-figure rule. A1(a)(ii)-(iv), A2(b), and A2(c) all carry numerical results forward from an earlier step — an answer that applies the correct method consistently from an earlier arithmetic slip still earns the marks available for the later steps.

Consistency with other sittings. A2(a) (fed funds rate/OMO), A3(b) (conventional vs. unconventional tools), and parts of A3(a) (accelerationist Phillips curve, Taylor rule) draw on the same shared theory library as the Winter 2025 EC3213 scheme, so the two never contradict each other. A2(b)'s covered interest arbitrage uses the same spot/forward quotes as the Winter 2025 sitting but a different interest-rate differential ($r\text{-pound} = 5\%$, not 6%) — worth noting if you're revising both sittings together, since the arbitrage *direction* is the same but the *profit* is smaller here.

Source basis. A1(b)'s PPP short-run/long-run evidence follows the module's own recommended reading (Abuaf and Jorion, "Purchasing Power Parity in the Long Run," Journal of Finance, 1990) and lecture material stating that PPP fits the data poorly over short intervals. All bond-valuation and compounding figures were independently verified before being written into the scheme.

Section A · Question 1

Bond valuation and interest rate risk, PPP over time, and the money multiplier — 100 marks

A1(a)

[45 marks]

Assume that investors require a rate of return for a given bond of 10% with a five-year maturity. The bond has a par value of €1,000 and has annual interest payments of €100. (i) Calculate the value of the bond. (ii) How much would the value of the bond change if the required rate of return increases by 2%? (iii) How would the bond value change if the required rate of return decreases by 2%? (iv) Illustrate and explain the relationship between the investors' required rate of return and the value of a bond using the figures calculated in (a)(i), (a)(ii) and (a)(iii). Explain the implications of your answer in terms of interest rate risk, premium bonds, and discount bonds.

A1(b)

[35 marks]

Does empirical evidence suggest PPP may be more likely to hold in the short-run or the long-run? Can you suggest an explanation for why this might be true?

A1(c)

[20 marks]

With the aid of an example, explain the money multiplier.

FULL MODEL ANSWER

(i) Setting up the valuation: a 5-year bond with a €100 annual coupon (a 10% coupon rate on €1,000 par) and a 10% required return, using $P = C \times [1 - (1+r)^{-n}] / r + \text{Par} / (1+r)^n$ from the exam's own formula sheet.

Year (t)	Cash flow (€)	Discount factor (1.10) ^{-t}	PV of cash flow (€)
1	100.00	0.9091	90.91

2	100.00	0.8264	82.64
3	100.00	0.7513	75.13
4	100.00	0.6830	68.30
5	1,100.00 (coupon + par)	0.6209	683.01

Bond price	€1,000.00
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Result: the bond prices **exactly at par** (€1,000.00) — as with any bond, this is not a coincidence: whenever the coupon rate (here, €100/€1,000 = 10%) equals the required rate of return, the bond always prices at par.

(ii)-(iii) Repricing at 12% and 8%: the same cash flows, discounted at the two new required rates.

Required return	Bond price	Change from base (€1,000.00)
8% (−2%)	€1,079.85	+€79.85 (+7.99%)
10% (base)	€1,000.00	—
12% (+2%)	€927.90	−€72.10 (−7.21%)

Bond Price-Yield Relationship (5-Year, €100 Annual Coupon)

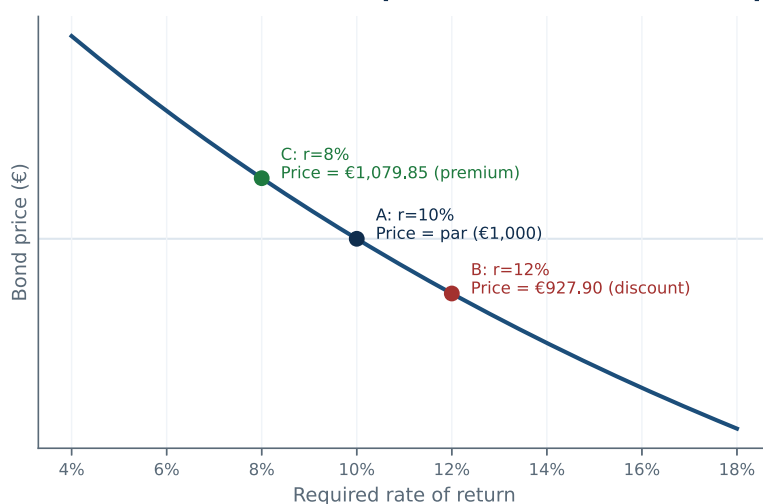


Fig. 1 — The bond price-yield relationship: price and required return move inversely, and the curve is convex — the price gain from a 2-point rate decrease (+€79.85) exceeds the price loss from an equal 2-point rate increase (−€72.10).

(iv) The relationship, illustrated: bond price and required return move **inversely** — as the required return rises, the fixed future cash flows are discounted more heavily and the price falls; as it falls, the price rises. The curve relating the two is not a straight line but **convex**: notice that the €2-point rate *decrease* increased the price by €79.85, while the equal €2-point rate *increase* decreased the price by only €72.10 — the same size rate move produces a larger price effect when rates fall than when they rise.

Implications

- **Interest rate risk:** because bond prices move inversely with required returns, any bondholder who might need to sell before maturity is exposed to **interest rate risk** — a rise in market rates after purchase forces a capital loss if sold early. The convexity result above adds a further nuance: the downside from a rate *increase* is smaller in magnitude than the upside from an equal rate

decrease, so interest rate risk is asymmetric in the bondholder's favour for a given-sized rate move in either direction — small comfort in a rising-rate environment, but a real feature of fixed-income risk.

- **Discount bonds:** when the required return exceeds the coupon rate (here, at 12% vs. the 10% coupon), the bond prices **below par** (€927.90) — investors demand a discount to compensate for a coupon that's now below the going market rate.
- **Premium bonds:** when the required return is below the coupon rate (here, at 8% vs. the 10% coupon), the bond prices **above par** (€1,079.85) — investors are willing to pay a premium to lock in a coupon that beats the going market rate.

Mark Allocation — Part (a) (45 marks)

Tier	Marks	What separates this tier
35-45	35-45	All three bond prices correctly calculated AND the inverse, convex price-yield relationship correctly illustrated and explained AND all three implications (interest rate risk, premium bonds, discount bonds) correctly explained with reference to the specific figures calculated.
23-34	23-34	All three prices correctly calculated, with the inverse relationship stated, but the convexity point is missed and/or the three implications are only partially explained.
12-22	12-22	Only the base-case (10%) price correctly calculated; the $\pm 2\%$ repricing has errors, or the implications are generic and not tied to the actual figures.
0-11	0-11	No coherent bond valuation attempted.

Long-run, not short-run: empirical evidence — including the module's own recommended reading, Abuaf and Jorion's "Purchasing Power Parity in the Long Run" (1990) — finds that PPP **fits the data poorly over short intervals of time**, but performs considerably better as a long-run equilibrium relationship: actual exchange rates deviate from their PPP-implied levels for extended periods, but tend to revert slowly back toward those levels over a horizon of several years.

Why this might be true: PPP is fundamentally a **goods-market arbitrage** condition — it holds because persistent price differences get competed away by trade. But in the **short run**, the exchange rate is also influenced by a range of factors that have nothing to do with goods-market arbitrage and can dominate it entirely: capital flows chasing interest-rate differentials (see Q. A2(b)), shifts in risk appetite and speculative positioning, and news about future monetary and fiscal policy — all of which can move exchange rates by more, and faster, than the underlying inflation differentials PPP is built on. Goods prices themselves are also **sticky** in the short run (menu costs, existing contracts, gradual pass-through of exchange-rate changes into local prices), so even where an exchange-rate move should eventually trigger a goods-market response, that response takes time to materialise. Over the **long run**, these short-run financial-market and price-stickiness effects fade, trade and arbitrage pressures have time to operate fully, and the exchange rate converges back toward the level PPP would predict from relative price levels.

EXAM TECHNIQUE

A strong answer doesn't just assert "short-run has more noise" — it identifies specific mechanisms (capital flows/interest differentials, price stickiness, speculative positioning) that operate on a different timescale than goods-market arbitrage, which is precisely what explains the empirical long-run/short-run split.

Mark Allocation — Part (b) (35 marks)

Tier	Marks	What separates this tier
27-35	27-35	Correctly states that PPP holds better in the long run than the short run, ideally with reference to the empirical literature AND at least two specific mechanisms explaining why (capital flows/interest differentials, price stickiness, speculative factors) correctly explained.
18-26	18-26	Correctly states the long-run/short-run finding, with at least one mechanism given but not fully explained.
9-17	9-17	States that PPP "doesn't always hold" without a clear long-run/short-run distinction or mechanism.
0-8	0-8	No credible answer given, or the direction is reversed (claiming PPP holds better short-run).

The money multiplier: the ratio by which the money supply expands relative to an initial injection of reserves into the banking system, since banks re-lend most of any new deposit rather than holding it all as reserves, and each re-lent amount becomes a new deposit at another bank, which again re-lends most of it, and so on.

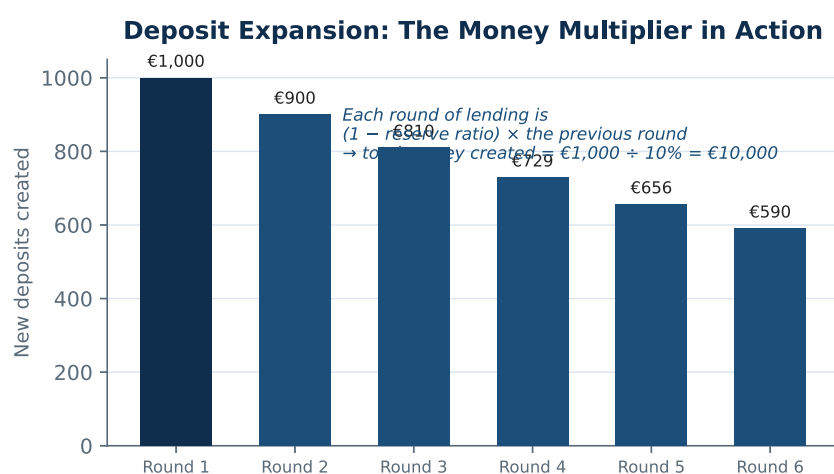


Fig. — Successive rounds of deposit expansion (illustrative: €1,000 initial deposit, 10% reserve ratio), shrinking geometrically and summing to the total money created.

Round	New deposit created (€)	Held as reserves (€)	Re-lent (€)
1	1,000.00	100.00	900.00
2	900.00	90.00	810.00
...	(geometric series continues)		
Total money created	€10,000.00		= €1,000 ÷ 10%

Result: with a reserve ratio of 10%, the simple money multiplier is $1 \div 10\% = 10$, so an initial €1,000 deposit ultimately supports €10,000 of money in the banking system. In practice, the actual (M1) multiplier is smaller than this simple reserve-ratio formula suggests, since it also depends on the public's currency-to-deposit ratio and banks' holding of excess reserves beyond the legal minimum.

Mark Allocation — Part (c) (20 marks)

Tier	Marks	What separates this tier
15-20	15-20	Correctly explains that banks re-lend deposits net of reserves, repeating across successive rounds AND gives a correct worked numeric example

with the multiplier formula ($1/\text{reserve ratio}$).

10-14	10-14	Correctly explains one round of lending increasing the money supply, but does not extend the reasoning to successive rounds or a full worked example.
5-9	5-9	Vague reference to "banks lending out deposits" without a reserve ratio or multiplier formula.
0-4	0-4	No credible explanation given.

Question A1 Total: 100 Marks

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Q1

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Section A · Question 2

Open market sales and the fed funds rate, covered interest arbitrage, and compounding — 100 marks

A2(a)

[40 marks]

Explain, with the aid of a diagram, how the Federal Funds interest rate is affected by an Open Market Sale by the Federal Reserve Bank. Illustrate how Open Market Operations are an effective tool for targeting the Fed Funds Rate.

A2(b)

[40 marks]

You have been given the following information: $r_{\$} = 4\%$ (annual, 3-month US-dollar commercial paper), $r_{£} = 5\%$ (annual, 3-month British-pound commercial paper), $S(\$ / £) = 1.5000$, $F_{1/4}(\$ / £) = 1.4988$. On the basis of the precise criteria: (i) In which commercial paper would you invest? (ii) In which currency would you borrow? (iii) How would you arbitrage? (iv) What is the profit from interest arbitrage per dollar borrowed?

A2(c)

[20 marks]

(i) You have €100,000 to invest now. The interest rate is 15% compounded annually. How long will it take for your investment to be worth €300,000? (ii) Calculate the future value of an initial investment of €8,000 after 3 years at an interest rate of 6% per annum if the compounding frequency is monthly.

FULL MODEL ANSWER

The market for reserves: the federal funds rate is the price that clears the **market for bank reserves** — determined by the demand banks have for holding reserves (partly to meet requirements, partly as a cushion against

unexpected deposit outflows) and the supply of reserves the Fed makes available.

The demand curve: downward-sloping in the fed funds rate — as i_{ff} falls, the opportunity cost of holding excess reserves (relative to lending them out) falls, so banks want to hold more of them. Since 2008 the Fed has paid interest on reserves (i_{or}) at a fixed spread below the fed funds target, so the demand curve **flattens out (becomes infinitely elastic) once i_{ff} falls to i_{or}** — below that point, banks would rather hold reserves than lend at a rate no better than what the Fed itself pays.

The supply curve: made up of non-borrowed reserves (supplied via OMOs) plus borrowed reserves (banks borrowing directly from the Fed at the discount rate, i_d). If $i_{ff} < i_d$, no bank will borrow from the Fed when cheaper funding is available in the interbank market, so borrowed reserves are zero and supply is **vertical**. If i_{ff} rises above i_d , banks profitably borrow from the Fed and re-lend at i_{ff} , so supply becomes **horizontal (perfectly elastic) at i_d** .

The Market for Reserves: Effect of an Open Market Sale

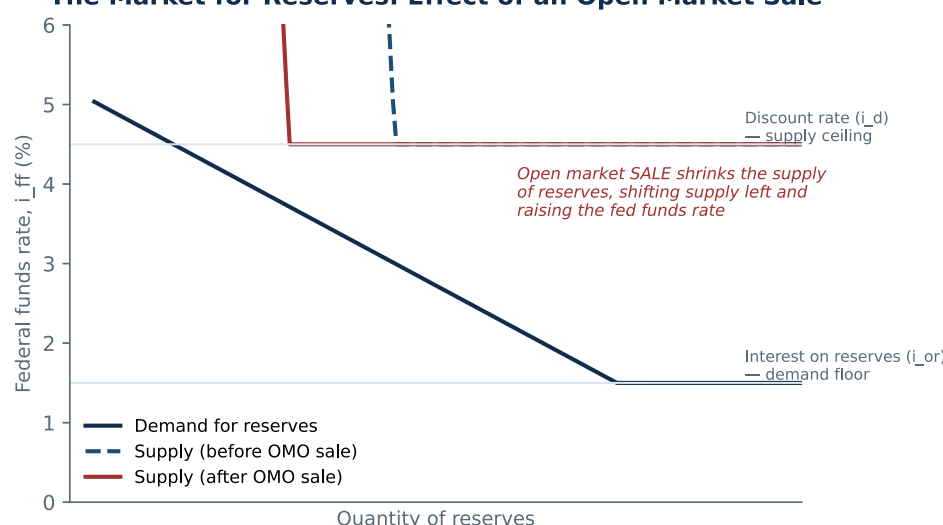


Fig. — An open market sale shrinks the supply of reserves, shifting the (vertical section of the) supply curve left. Where the new equilibrium remains on the downward-sloped section of the demand curve, this raises the federal funds rate from A to B.

Effect of an open market sale: an OMO sale drains reserves from the banking system, shifting the vertical portion of the supply curve to the **left**. Provided the new intersection with demand still falls on the downward-sloped section of the demand curve (i.e. the fed funds rate is still above i_{or}), this **raises the federal funds rate**. Conversely, an open market purchase shifts supply right and lowers i_{ff} .

Why OMOs are an effective targeting tool: OMOs give the Fed **complete control over the volume** transacted (unlike discount lending, where the bank itself decides whether to borrow), are **flexible and precise** — any desired quantity of reserves can be added or drained — and can be **quickly implemented and easily reversed** if conditions change, which is precisely what makes OMOs the Fed's primary conventional tool rather than the discount rate or reserve requirements.

EXAM TECHNIQUE

A common exam slip is describing the effect of an OMO sale without specifying *where* the new equilibrium falls relative to i_{or} — if reserves remain abundant enough that the intersection stays on the flat section of the demand curve at i_{or} , an OMO sale of typical size has **no effect** on the fed funds rate at all. The diagram's shape, not just the direction of the shift, is what determines the answer.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Demand and supply curves in the market for reserves correctly shaped (demand downward-sloping then flat at i_{or} ; supply vertical then flat at i_d) AND an OMO sale correctly shown shifting supply left, raising i_{ff} on the relevant section AND at least two specific reasons given for why OMOs are an effective targeting tool.
21-30	21-30	Correct diagram shape and correct direction of the OMO sale's effect, but the reasons OMOs are an effective tool are generic or missing.
11-20	11-20	Diagram shape roughly correct but the flat sections (at i_{or} and i_d) are missing or mislabelled, or the direction of the OMO sale's effect is reversed.
0-10	0-10	No credible diagram or explanation given.

Setting up the comparison: with a 3-month ($T = 0.25$ year) horizon, compare (A) investing \$1 in US commercial paper at 4% p.a., uncovered, against (B) converting \$1 to £ at spot, investing in British-pound commercial paper at 5% p.a. (note: 5% on this paper, not 6% as in the Winter 2025 sitting — check which rate you've been given), and selling the £ proceeds forward at $F_{1/4}$ to lock in the \$ conversion.

Strategy	Calculation	Result per \$1
(A) Invest \$1 in US CP	$1 \times (1 + 0.04 \times 0.25)$	\$1.010000
Convert \$1 to £ at spot	$1 \div 1.5000$	£0.666667
Invest £0.666667 in UK CP at 5%	$0.666667 \times (1 + 0.05 \times 0.25)$	£0.675000
(B) Convert back at $F_{1/4} = 1.4988$	0.675000×1.4988	\$1.011690

(i) Which commercial paper to invest in: the covered British-pound strategy (B) returns **\$1.011690** per dollar, more than the US strategy (A)'s \$1.010000 — so invest in **British-pound commercial paper**, covered forward.

(ii) Which currency to borrow: borrow in **US dollars** at 4% — the currency you are not investing in — to fund a self-financing, zero-net-investment arbitrage.

(iii) How to arbitrage: borrow \$1 at $r\$ = 4\%$; convert immediately to £ at spot ($S = 1.5000$); invest the £ proceeds in British-pound commercial paper at $r£ = 5\%$; simultaneously sell the maturing £ proceeds forward 3 months at $F_{1/4} = 1.4988$ to lock in the \$ conversion today; at maturity, convert back to \$ at the contracted forward rate, repay the \$ loan plus interest, and keep the difference — fully covered, with no exchange-rate risk.

(iv) Profit from interest arbitrage per dollar borrowed: \$1.011690 (covered £ strategy) – \$1.010000 (cost of repaying the \$ loan) = **\$0.001690 profit per dollar borrowed**, or 0.1690% over the 3-month period.

EXAM TECHNIQUE

Sanity check: the CIP-consistent forward here is $F = S \times (1+r\$T)/(1+r£T) = 1.5000 \times$

1.01/1.0125 $\approx$ **1.4963**. The actual quoted forward (1.4988) is again higher than this — the same direction of mispricing as the Winter 2025 sitting — but the gap between the actual and CIP-implied forward is narrower here (since $r_{\text{£}}$ is only 1 point above $r_{\text{\$}}$ this time, not 2), which is exactly why the arbitrage profit (\$0.001690) is smaller than in the Winter 2025 sitting (\$0.004188) despite an identical spot/forward quote. Don't assume the answer from one sitting carries over to the other without redoing the arithmetic against the actual $r_{\text{£}}$ given.

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly computes both the uncovered US return and the covered £ return using $r_{\text{£}}=5\%$ AND correctly identifies which CP to invest in and which currency to borrow AND correctly describes the arbitrage mechanics AND arrives at \$0.001690 (or a consistent own-figure result).
21-30	21-30	Both returns correctly calculated and the direction of arbitrage correctly identified, but the arbitrage mechanics description is incomplete or the profit figure has an arithmetic error.
11-20	11-20	Only one of the two strategies' returns correctly calculated, or the direction of arbitrage is reversed.
0-10	0-10	No coherent covered-interest-arbitrage calculation attempted.

(i) Time for €100,000 to grow to €300,000 at 15% p.a.: solving $FV = PV(1+r)^n$ for n : $300,000 = 100,000 \times 1.15^n$, so $1.15^n = 3$, and $n = \ln(3) / \ln(1.15)$.

Step	Value
FV / PV	3.0000
$\ln(3)$	1.0986
$\ln(1.15)$	0.1398
$n = \ln(3) / \ln(1.15)$	≈ 7.86 years

(ii) FV of €8,000 after 3 years at 6% p.a., monthly compounding: using $FV = PV(1 + k/m)^{mn}$ from the exam's own formula sheet, with $k=6\%$, $m=12$ (monthly), $n=3$ years.

Input	Value
PV	€8,000
k/m (monthly rate)	$6\% \div 12 = 0.5\%$
mn (number of periods)	$12 \times 3 = 36$
$FV = 8,000 \times (1.005)^{36}$	€9,573.44

EXAM TECHNIQUE

A common slip on (ii) is using the annual rate (6%) directly over 3 periods instead of converting to a monthly rate (0.5%) over 36 periods — since the question specifies monthly compounding, both the rate and the period count must be adjusted, exactly as in the semi-annual bond-pricing questions on other EC3005/EC3008 papers.

Mark Allocation — Part (c) (20 marks)

Tier	Marks	What separates this tier
15-20	15-20	Both (i) and (ii) correctly set up and calculated, with the correct formula

applied in each case (solving for n via logarithms in (i); correct monthly-compounding conversion in (ii)).

10-14	10-14	Both correctly set up, but one has an arithmetic error, or (ii) uses the annual rate/period incorrectly.
5-9	5-9	Only one of the two parts correctly calculated.
0-4	0-4	Neither part correctly calculated.

Question A2 Total: 100 Marks

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Q2

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Section A · Question 3

The Phillips curve, the Taylor rule, monetary policy tools, and covered interest parity — 100 marks

A3(a)

[40 marks]

Outline the economic reasoning behind: (i) The Phillips Curve; (ii) The Accelerationist Phillips Curve; and (iii) The Taylor Rule.

A3(b)

[30 marks]

Briefly outline the difference between conventional and unconventional monetary policy tools.

A3(c)

[30 marks]

'The forward premium (discount) of a foreign currency should be equal to the interest rate differential between the domestic and foreign currencies/countries in order to prevent arbitrage.' Do you agree? Why?

FULL MODEL ANSWER

(i) The (traditional) Phillips Curve — the reasoning: A.W. Phillips's 1958 LSE study documented a **negative statistical relationship** between wage inflation and unemployment in the UK — tight labour markets were associated with higher wage inflation, and vice versa; Solow and Samuelson (1960) replicated the finding for the US using price inflation. The economic mechanism behind it: lower unemployment increases workers' bargaining power, leading to higher nominal wage demands, which feed through into higher prices — a wage-price spiral. The original curve was read as a genuine, stable **policy trade-off**: a central bank could apparently buy lower unemployment at the cost of somewhat higher, but stable, inflation, and this was the dominant view

(ii) The Accelerationist Phillips Curve — the reasoning: Friedman argued workers actually bargain over **real**, not nominal, wages, and reasoned about how inflationary expectations are formed — specifically, that they are **adaptive**: this year's expected inflation is simply last year's actual inflation ($\pi_t^e = \pi_{t-1}$). Substituting this into the expectations-augmented Phillips curve gives $\pi_t = \pi_{t-1} - \gamma(U_t - U^*)$: it is the *change* in inflation, not its level, that relates to the unemployment gap. This means a central bank *can* hold unemployment below its natural rate U^* , but only by accepting **ever-accelerating** inflation — each period's attempt to exploit the trade-off pushes expected inflation up further, shifting the short-run curve outward the following period, with no stable inflation level consistent with $U < U^*$ forever.

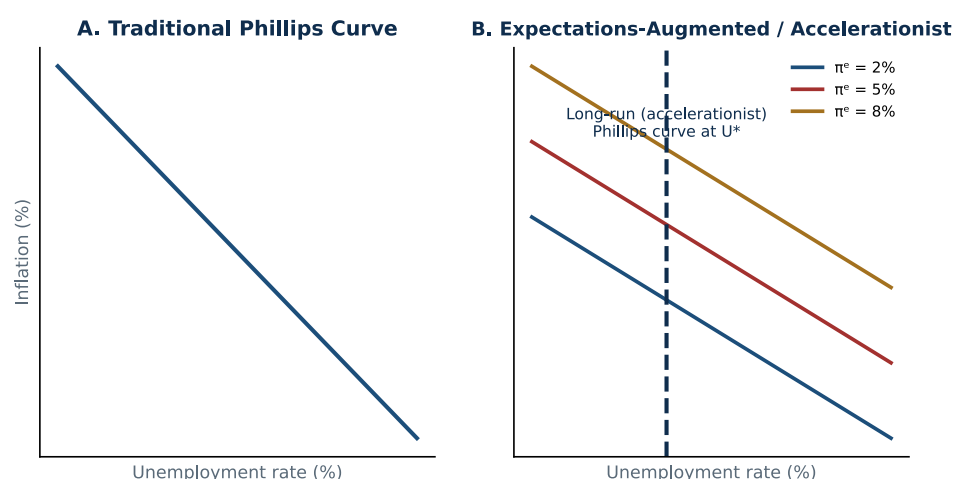


Fig. 1 — The traditional Phillips curve (A) suggests a stable trade-off; the expectations-augmented/accelerationist view (B) shows the curve shifting outward as expected inflation rises, with only a vertical long-run curve at the natural rate U^* .

Why Friedman's critique was vindicated: loose US fiscal and monetary policy in the 1960s initially produced exactly the traditional trade-off (inflation rose, unemployment fell), but as the public's expectations adjusted to the new, higher inflation, the trade-off worsened through the late 1960s, and by the 1970s produced **stagflation** — rising inflation *and* rising unemployment together, a positive correlation the original Phillips curve couldn't explain at all, but which the accelerationist model predicts exactly.

(iii) The Taylor Rule — the reasoning: a simple, transparent rule prescribing how a central bank's policy rate should respond to observable economic conditions: $i_{ff} = \pi + r^* + \frac{1}{2}(\pi - \pi^*) + \frac{1}{2}(\text{output gap})$. Including *both* an inflation gap and an output gap reflects that a central bank cares not only about keeping inflation near target, but also about smoothing business-cycle fluctuations of output around its potential — a compact summary of a central bank pursuing both objectives simultaneously, and widely used as a benchmark for assessing whether actual policy looks "too loose" or "too tight" relative to what observable conditions would prescribe.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	The traditional Phillips curve's negative-relationship finding and wage-price-spiral mechanism correctly explained AND the accelerationist curve correctly derived from adaptive expectations with the $\pi_t = \pi_{t-1} - \gamma(U_t - U^*)$ relationship stated AND the Taylor rule formula correctly stated with the dual-mandate rationale explained.
21-30	21-30	All three correctly stated, but the underlying reasoning for at least one

(why the original trade-off broke down; why the rule includes both gaps) is not explained.

11-20 **11-20** Only two of the three correctly explained.

0-10 **0-10** Only one, or none, correctly explained.

Conventional tools: the three tools central banks use in **normal times** to control the money supply and interest rates — **open market operations**, **discount lending**, and **reserve requirements**. All three work by adjusting the quantity of reserves in the banking system, and are well-understood, precise, and easily reversible.

Unconventional tools: measures developed specifically because, during the 2008 financial crisis, conventional tools proved insufficient once short-term interest rates hit the **zero lower bound** — there was no more room to cut a policy rate that was already near zero. Unconventional tools instead **directly target the cost and availability of external finance** to banks, households and firms — through central-bank liquidity provision, large-scale asset purchases (quantitative easing), and forward guidance (explicit communication about the likely future path of policy, intended to shape expectations directly rather than working solely through the current policy rate).

The key distinction: conventional tools work by moving the price of reserves (the fed funds rate) within a functioning market where that rate still has room to move; unconventional tools were developed for the situation where that room has run out, and instead act on the size and composition of the central bank's own balance sheet, or on market expectations directly, to keep providing monetary accommodation even at the zero lower bound.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Conventional tools correctly named (OMO, discount lending, reserve requirements) AND unconventional tools correctly named (QE, forward guidance) AND the zero-lower-bound rationale for why unconventional tools were developed is explained.
15-22	15-22	Both categories of tool correctly named, but the zero-lower-bound rationale is missing.
8-14	8-14	Only one category correctly named.
0-7	0-7	No credible distinction given.

Agree — this is exactly the (approximate) covered interest parity condition.

Why this is correct: the statement is essentially the **approximate form of covered interest parity (CIP)**: $(F-S)/S \approx r_{\text{domestic}} - r_{\text{foreign}}$, where the forward premium (or discount) on the left roughly equals the interest-rate differential on the right. The exact (non-approximated) version, used throughout Q. A2(b), is $F = S \times (1+r_{\text{domestic}} \times T)/(1+r_{\text{foreign}} \times T)$ — the approximation drops second-order cross terms that are small for short horizons and typical interest rate levels.

Why it must hold to prevent arbitrage: Q. A2(b) demonstrates the mechanism directly — whenever the quoted forward premium/discount deviates from the interest rate differential, a riskless, self-financing covered arbitrage becomes available: borrow in the currency with the lower covered return, invest in the currency with the higher covered return, and lock in the

conversion with a forward contract, exactly as in that question. This is a genuine no-arbitrage condition (unlike uncovered interest parity, which relies on risk-neutral expectations and does not hold as reliably in practice) — because every leg of the covered trade is contracted today, with no exchange-rate risk left over, any deviation represents a risk-free profit opportunity that well-capitalised arbitrageurs would exploit until it closed.

The precise direction: specifically, the currency with the **higher** interest rate must trade at a forward **discount** (F below S, if it's the foreign currency being quoted per unit of domestic currency) of a size that exactly offsets its interest rate advantage — otherwise, investing in the higher-rate currency covered would strictly dominate investing in the lower-rate currency, and the resulting capital flows (borrowing the low-rate currency, buying the high-rate currency and its forward cover) would themselves move the spot rate, the forward rate, or both interest rates until the condition was restored.

EXAM TECHNIQUE

This question is testing the same result as A2(b) from the opposite direction — A2(b) gives you a specific numerical case where the condition is *violated* and asks you to exploit it; A3(c) asks you to state and justify the general condition itself. A strong answer references this connection explicitly rather than treating the two as unrelated.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly agrees and identifies the statement as (approximate) CIP AND correctly explains the no-arbitrage mechanism (covered borrowing/investing across currencies) that enforces it AND correctly states which currency (higher-rate) must trade at a forward discount.
15-22	15-22	Correctly agrees and identifies the condition as CIP, with a general no-arbitrage justification, but the specific direction (which currency trades at a discount) is not stated or is incorrect.
8-14	8-14	Agrees with the statement but without a clear justification beyond restating it.
0-7	0-7	Disagrees without valid justification, or no credible answer given.

Question A3 Total: 100 Marks